

## Surface Integrals: Two Flavors in Parallel

In Section 16.2 we defined **two kinds** of line integrals — one for scalar functions, one for vector fields. Now we lift each of them up a dimension.

|                                     | Scalar version         | Vector version                                                         |
|-------------------------------------|------------------------|------------------------------------------------------------------------|
| <b>Line integral</b> (1D object)    | $\int_C f \, ds$       | $\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \vec{T} \, ds$   |
| <i>weighted by</i>                  | arc length $ds$        | <b>tangent</b> component $\vec{F} \cdot \vec{T}$                       |
| <i>measures</i>                     | total of $f$ along $C$ | flow <b>along</b> $C$ (work)                                           |
| <b>Surface integral</b> (2D object) | $\iint_S f \, dS$      | $\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n} \, dS$ |
| <i>weighted by</i>                  | surface area $dS$      | <b>normal</b> component $\vec{F} \cdot \vec{n}$                        |
| <i>measures</i>                     | total of $f$ over $S$  | flow <b>through</b> $S$ ( <b>flux</b> )                                |

**The shift:** for curves, “flow” meant *tangential* (along the curve). For surfaces, “flow” means *perpendicular* (through the surface). That’s why flux uses  $\vec{n}$  instead of  $\vec{T}$ .

Each line integral from 16.2 has a surface-integral twin. Scalar  $\leftrightarrow$  scalar ( $ds \rightarrow dS$ ); vector (work)  $\leftrightarrow$  vector (flux) ( $\vec{T} \rightarrow \vec{n}$ ).

## Scalar Surface Integrals: The Idea

Imagine a thin metal sheet shaped like a surface  $S$ , with density  $\rho(x, y, z)$  at each point. The **total mass** is:

$$\text{mass} = \iint_S \rho(x, y, z) \, dS$$

More generally,  $\iint_S f(x, y, z) \, dS$  adds up the values of  $f$  over the surface, weighted by area.

$\iint_S f \, dS$  is the surface analog of  $\int_C f \, ds$ . Replace “arc-length element  $ds$ ” with “surface-area element  $dS$ ” — everything else is the same.

## Computing $\iint_S f \, dS$ from a Parametrization

From Section 16.6, if  $S$  is parametrized by  $\vec{r}(u, v)$  over domain  $D$ , then

$$dS = |\vec{r}_u \times \vec{r}_v| \, dA.$$

### Scalar Surface Integral (Parametric Form)

$$\iint_S f(x, y, z) \, dS = \iint_D f(\vec{r}(u, v)) |\vec{r}_u \times \vec{r}_v| \, dA$$

Recipe: (1) parametrize  $S$  as  $\vec{r}(u, v)$ , (2) compute  $|\vec{r}_u \times \vec{r}_v|$ , (3) substitute into a double integral over  $D$ . Exactly like a line integral, one dimension up.

## Special Case: Surface Integral over a Graph $z = g(x, y)$

When  $S$  is a graph  $z = g(x, y)$ , parametrize with  $x$  and  $y$  directly:  $\vec{r}(x, y) = \langle x, y, g(x, y) \rangle$ .

**Recall from 16.6:**

$$|\vec{r}_x \times \vec{r}_y| = \sqrt{g_x^2 + g_y^2 + 1}$$

So the scalar surface integral becomes:

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{g_x^2 + g_y^2 + 1} dA$$

where  $D$  is the projection of  $S$  onto the  $xy$ -plane.

If  $f \equiv 1$ , this reduces to the **surface-area formula** from Section 15.6:

$$\text{Area}(S) = \iint_D \sqrt{g_x^2 + g_y^2 + 1} dA$$

Scalar surface integrals include surface area as a special case — just as  $\int_C 1 ds$  gives arc length.

For graphs  $z = g(x, y)$ : replace  $dS$  with  $\sqrt{g_x^2 + g_y^2 + 1} dA$  and substitute  $z = g(x, y)$  into  $f$ . Most textbook problems fit this shape.

### Example 1: Scalar Surface Integral over a Graph

#### Example 1

Evaluate  $\iint_S y dS$ , where  $S$  is the surface  $z = x + y^2$  over the rectangle  $0 \leq x \leq 1, 0 \leq y \leq 2$ .

*Identify  $g(x, y)$ , compute  $g_x$  and  $g_y$ , then set up the integral using the graph formula.*

### Quick Check: Set Up a Scalar Surface Integral

Set up (but do not evaluate)  $\iint_S (x^2 + z) dS$  where  $S$  is the part of the plane  $z = 4 - 2x$  over the rectangle  $0 \leq x \leq 2, 0 \leq y \leq 3$ .

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Find  $g(x, y)$ , compute  $g_x$  and  $g_y$ , write  $\sqrt{g_x^2 + g_y^2 + 1}$ , substitute  $z = g(x, y)$  into  $f$ , then write the double integral.

## Example 2: Scalar Surface Integral on a Sphere

### Example 2

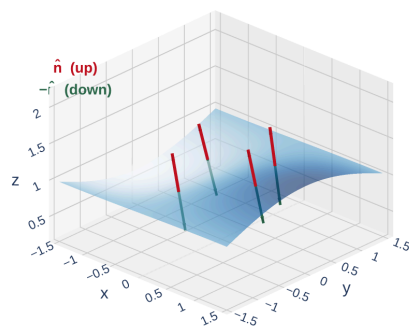
Evaluate  $\iint_S y^2 dS$  where  $S$  is the part of the sphere  $x^2 + y^2 + z^2 = 4$  that lies inside the cylinder  $x^2 + y^2 = 1$  and above the  $xy$ -plane.

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Parametrize the sphere using spherical coordinates with  $\rho = 2$ :  $\vec{r}(\theta, \phi) = \langle 2 \sin \phi \cos \theta, 2 \sin \phi \sin \theta, 2 \cos \phi \rangle$ . What are the bounds on  $\phi$ ?

## Oriented Surfaces (Bridge to Flux)

Before we can define flux, we need to pick a **direction** at every point of the surface — an orientation.

Orientable surface (two choices of  $\hat{n}$ )

Möbius strip (non-orientable)



[Interactive version](#)

An **orientable surface** has two distinct sides. At each point there are two unit normals,  $\vec{n}$  and  $-\vec{n}$ . Choosing one consistently over all of  $S$  gives an **orientation**.

**Standard conventions:**

- **Closed surfaces** (sphere, torus, cube): *outward* normal is positive.
- **Graphs**  $z = g(x, y)$ : *upward* normal is positive.
- Every problem statement tells you which orientation to use.

Not every surface is orientable — the **Möbius strip** has only one side. But every surface we will integrate over in this course is orientable.

An orientation is a consistent choice of unit normal. For flux integrals the orientation *determines the sign* of the answer: reverse the orientation and the flux changes sign.

## Unit Normals for the Two Most Common Surfaces

For flux we need the unit normal  $\vec{n}$ . Two parametrizations cover almost every problem.

**Graph**  $z = g(x, y)$ ,  $\vec{r}(x, y) = \langle x, y, g(x, y) \rangle$ :

$$\vec{r}_x \times \vec{r}_y = \langle -g_x, -g_y, 1 \rangle$$

$$\vec{n}_{\text{up}} = \frac{\langle -g_x, -g_y, 1 \rangle}{\sqrt{g_x^2 + g_y^2 + 1}}$$

The  $+1$  in the third slot means this vector points upward (so it *is* the positive orientation).

**Sphere** of radius  $R$ , spherical parametrization.

First, the **easy-to-remember outward unit normal**:

$$\vec{n}_{\text{out}} = \frac{\langle x, y, z \rangle}{R} = \hat{r}$$

(just the position vector, normalized.) Then the cross product factors cleanly as

$$\vec{r}_\phi \times \vec{r}_\theta = R^2 \sin \phi \hat{r}.$$

Since  $\sin \phi \geq 0$ , this points outward automatically.

To **reverse** either orientation, multiply by  $-1$  (or swap the order of the cross product).

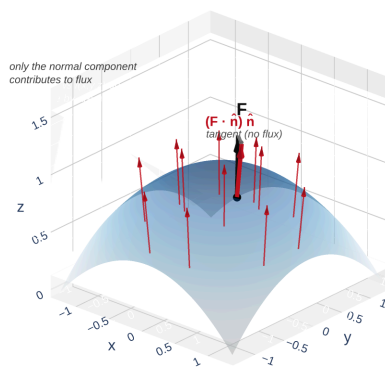
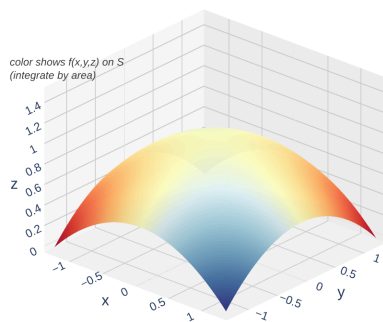
Graphs:  $\vec{r}_x \times \vec{r}_y = \langle -g_x, -g_y, 1 \rangle$  points up. Spheres:  $\vec{r}_\phi \times \vec{r}_\theta$  points out. Both match the standard positive orientation.

## Flux: Definition and Parametric Formula

Now the big idea: **how much** of a vector field  $\vec{F}$  flows *through* a surface.

Scalar surface integral:  $\iint_S f \, dS$

Flux:  $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$



[Interactive version](#)

At each point, only the piece of  $\vec{F}$  **perpendicular** to  $S$  passes through — that piece is  $\vec{F} \cdot \vec{n}$ .

**Flux of  $\vec{F}$  across an Oriented Surface  $S$**

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n} \, dS$$

**Parametric formula.** Since  $\vec{n} \, dS = (\vec{r}_u \times \vec{r}_v) \, dA$ , the  $|\cdot|$  on the cross product cancels the  $|\cdot|$  in the denominator of  $\vec{n}$ :

**Theorem: Flux via a Parametrization**

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA$$

Use the cross product *itself* — not its magnitude. Order matters:  $\vec{r}_u \times \vec{r}_v$  vs.  $\vec{r}_v \times \vec{r}_u$  gives opposite orientations.

Flux measures flow **through**  $S$  the way work measures flow **along**  $C$ . Dot  $\vec{F}$  with  $\vec{r}_u \times \vec{r}_v$  (no magnitude, no square root).

**Flux through a Graph  $z = g(x, y)$**

Plug  $\vec{r}_x \times \vec{r}_y = \langle -g_x, -g_y, 1 \rangle$  into the parametric flux formula.

If  $\vec{F} = \langle P, Q, R \rangle$ , then

$$\vec{F} \cdot (\vec{r}_x \times \vec{r}_y) = -P g_x - Q g_y + R$$

(with  $P, Q, R$  evaluated at  $(x, y, g(x, y))$ ).

**Upward orientation** (positive):

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D (-P g_x - Q g_y + R) \, dA.$$

**Downward orientation:** flip the sign.

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D (P g_x + Q g_y - R) \, dA.$$

**No square root.** The magnitude from  $dS$  cancels the denominator of  $\vec{n}$ . This is the single most useful formula in the section.

Flux through a graph:  $\iint_D (-Pg_x - Qg_y + R) dA$  (upward). Flip sign for downward. Compare with the scalar version, which *did* need  $\sqrt{g_x^2 + g_y^2 + 1}$ .

### Quick Check: Identify the Flux Setup

$\vec{F}(x, y, z) = \langle 0, 0, z \rangle$  and  $S$  is the part of the paraboloid  $z = 1 - x^2 - y^2$  above the  $xy$ -plane, with upward orientation. Write  $\vec{F} \cdot (\vec{r}_x \times \vec{r}_y)$  as a function of  $x$  and  $y$ .

Find  $g(x, y)$ , form  $\langle -g_x, -g_y, 1 \rangle$ , substitute  $z = g(x, y)$  into  $\vec{F}$ , then dot.

### Example 3: Flux across a Plane

#### Example 3

Find the flux of  $\vec{F}(x, y, z) = \langle x, 2y, 3z \rangle$  upward through the triangular portion of the plane  $2x + y + 2z = 6$  in the first octant.

Solve for  $z = g(x, y)$ , find  $g_x$  and  $g_y$ , apply the upward-graph flux formula, and find the domain in the  $xy$ -plane.

## Example 4: Flux across the Unit Sphere

### Example 4

Find the flux of  $\vec{F}(x, y, z) = \langle z, y, x \rangle$  outward across the unit sphere  $x^2 + y^2 + z^2 = 1$ .

Use spherical parametrization with  $R = 1$ . Recall  $\vec{r}_\phi \times \vec{r}_\theta = \sin \phi \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle$  points outward.

## Example 5: Flux on a Cone (Downward Orientation)

### Example 5

Evaluate  $\iint_S \vec{F} \cdot d\vec{S}$  where  $\vec{F}(x, y, z) = \langle -x, -y, z^3 \rangle$  and  $S$  is the part of the cone  $z = \sqrt{x^2 + y^2}$  between  $z = 1$  and  $z = 3$ , with **downward** orientation.

Use  $g(x, y) = \sqrt{x^2 + y^2}$  and the downward graph formula  $\iint_D (Pg_x + Qg_y - R) dA$ . The domain is the annulus  $1 \leq r \leq 3$  in polar.

## Quick Check: Does Orientation Matter?

If the flux of  $\vec{F}$  through a surface  $S$  with upward orientation is 12, what is the flux with downward orientation?

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Think about how flipping  $\vec{n}$  changes the flux integrand.

### Application: Heat Flow across a Surface

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If the temperature in a solid is  $u(x, y, z)$ , the **heat-flow vector** is

$$\vec{F} = -K \nabla u \quad (K > 0, \text{ thermal conductivity}).$$

Heat flows from hot to cold — opposite the gradient.

The **rate of heat flow** across a surface  $S$  is

$$\text{rate} = \iint_S \vec{F} \cdot d\vec{S} = -K \iint_S \nabla u \cdot d\vec{S}.$$

#### Example 6

Temperature inside a ball:  $u = C/r$  where  $r = \sqrt{x^2 + y^2 + z^2}$ . Find the rate of heat flow across the sphere  $S$  of radius  $a$ .

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Compute  $\nabla u$ , form  $\vec{F} = -K \nabla u$ , then dot with the outward normal  $\vec{n} = \frac{1}{a} \langle x, y, z \rangle$  on the sphere  $r = a$ .

### Homework

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**Section 16.7:** 5, 7, 9, 11, 13, 15, 17, 21, 23, 25, 27, 29, 43